Overlap Does Not Decide Independence

Topic 4.6 · TODAY'S PROBLEM

Suppose a school surveys 200 students about whether they own at least one pet and whether they play a musical instrument. Let P be the event that a randomly selected surveyed student owns a pet and I the event that the student plays an instrument.

Aaliyah: *The events overlap because 84 students do both. Since they are not mutually exclusive, they must be independent.*

Marcus: *Overlap does not decide independence; the relevant probabilities must be compared.*

Pet ownership and instrument playing

Pet?	Plays	Does not	Total
Yes	84	36	120
No	16	64	80
Total	100	100	200

FIRST TAKE (5 min, on your sheet)

Do the events P and I seem independent? Cite one count or comparison from the table.

DOK 1 **(a)** State the counts for P, I, and $P \cap I$.

DOK 2 **(b)** Compute $P(I)$, $P(I \mid P)$, $P(P \cap I)$, and $P(P \cup I)$.

DOK 3 **(c)** Decide whether pet ownership and instrument playing are independent for these students. Cite the two probabilities that settle it. Then evaluate Aaliyah's reasoning: explain how mutually exclusive and independent events differ, give a pair of events from this context that is mutually exclusive but not independent, and state what independence would have allowed you to compute with the multiplication rule.

Video + follow-along: `u4_lesson6_live.html` (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.